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Conditional Second-Order Generalized Estimating Equations for Generalized Linear and Nonlinear Mixed-Effects Models

Edward F. VONESH, Hao WANG, Lei NIE, and Dibyen MAJUMDAR

Generalized linear and nonlinear mixed-effects models are used extensively in health care research, including applications in pharmacokinetics, clinical trials, and epidemiology. Because the underlying model may be nonlinear in the random effects, there will generally be no closed-form expression for the marginal likelihood or indeed for the marginal moments. Consequently, estimation is often carried out either using numerical integration techniques or by approximating the marginal likelihood and/or marginal moments using first-order expansion methods. An advantage of the first-order methods is that they do not necessarily require specification of a conditional likelihood to estimate the regression parameters of interest. However, in many cases they may not take full advantage of the fact that the conditional variance depends on both fixed and random effects. In this article we propose using conditional second-order generalized estimating equations (CGEE2) to estimate both fixed- and random-effects parameters. Under mild regularity conditions, the CGEE2 estimator is shown to be consistent and asymptotically efficient with a rate of convergence depending on both the number of subjects and the number of observations per subject. We compare the CGEE2 estimator against alternative estimators using limited simulation and demonstrate its utility with a numerical example.

KEY WORDS: Asymptotics; First-order linearization; Gaussian random effects; Laplace approximation; Penalized extended least squares; Quadratic exponential family.

1. INTRODUCTION

In many applications, the analysis of clustered data including longitudinal data or repeated measurements requires that one estimate both fixed- and random-effects parameters from a generalized linear or nonlinear mixed-effects model. Estimation is complicated by the fact that these models are typically nonlinear with respect to the random effects and thus have no closed-form expressions for the marginal likelihood or indeed for the marginal moments. Various methods have been proposed to circumvent this problem, many of which have been summarized by Davidian and Giltinan (1995) and Vonesh and Chinchilli (1997). These include "exact" integration-based methods using either numerical integration techniques (e.g., Hedeker and Gibbons 1994; Liu and Pierce 1994; Diggle, Liang, and Zeger 1994; Pinheiro and Bates 1995), or Markov chain Monte Carlo techniques (e.g., Geweke 1989; Zeger and Karim 1991; Wakefield, Smith, Racine-Poon, and Gelfand 1994). Such integration-based techniques work extremely well when the number of random effects is small (e.g., one or two). However, because they are so computationally intensive, these techniques become increasingly more difficult to use as the number of random effects increases, particularly when the random effects are correlated. Moreover, their successful implementation via maximum likelihood requires complete specification of the relevant distributions.

As an alternative to numerical integration, two different first-order Taylor series expansion methods have been used to approximate the marginal likelihood function, marginal moments, or both. The first entails expanding the conditional mean or likelihood about 0, that is, the expectation of the random effects (e.g., Sheiner and Beal 1980; Goldstein 1991; Vonesh and Carter 1992; Longford 1994). Estimation is carried out by solving a set of first- or second-order generalized estimating equations (GEE1 or GEE2). The second method entails expanding about current estimates of the random effects (e.g., Stiratelli, Laird, and Ware 1984; Lindstrom and Bates 1990; Breslow and Clayton 1993; Wolfinger and O'Connell 1993). Estimation in this case is carried out by iteratively solving a set of conditional first-order generalized estimating equations (CGEE1). In many cases, CGEE1 can be derived via a Laplace approximation to an integrated likelihood, and the resulting estimator may be viewed as approximate maximum likelihood (Wolfinger 1993; Vonesh 1996). These first-order methods, particularly the conditional first-order method (i.e., CGEE1), appear to work quite well for continuous data, as is evident from the various simulation results in the literature (Davidian and Giltinan 1993; Vonesh 1992; Pinheiro and Bates 1995; Roe 1997; Wolfinger and Lin 1997). However, formal investigation into the asymptotic properties of these methods along with supporting simulation has been limited primarily to Gaussian-based models, where the conditional variance-covariance structure is independent of the random effects. Moreover, when the conditional distribution is from a quadratic exponential family, these conditional first-order methods fail to fully use information about fixed- and random-effects parameters that appear in the conditional variance.

In this article we consider estimating the parameters from a mixed-effects model using estimating equations involving both first- and second-order conditional moments. The proposed estimation scheme permits the modeling of multivariate continuous, discrete, or mixed responses by assuming that the conditional distribution of the response variable(s) belongs

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to a quadratic exponential family (e.g., Prentice and Zhao 1991). Joint estimation of fixed and random effects is achieved by alternately solving a set of conditional second-order generalized estimating equations (CGEE2) and estimating the variance-covariance parameters of the random effects using the Gaussian posterior approximation of Laird and Louis (1982). The entire approach is equivalent to linearizing the conditional first- and second-order moments about current estimates of the random effects and applying standard linear mixed-effects theory to update the model parameters. As such, this approach provides a natural extension to the conditional first-order methods described by Lindstrom and Bates (1990), Breslow and Clayton (1993), and others. The key difference is that by including conditional second-order moments in the set of estimating equations, CGEE2 can in some cases improve the efficiency of estimating both fixed- and random-effects parameters.

2. THE MODEL

Let $\boldsymbol{\beta}$ be an $s \times 1$ vector of fixed-effects parameters and let $\mathbf{b}_i$ be a $v \times 1$ vector of random-effects parameters unique to the ith cluster or subject ($i = 1, \ldots, n$). Let $\mathbf{x}_i$ and $\mathbf{z}_i$ be vectors of covariates linked to the fixed- and random-effects parameters. Given $\boldsymbol{\beta}$ and $\mathbf{b}_i$, let $\mathbf{y}_i = (y_{i1}, \ldots, y_{ip_i})'$ denote a $p_i \times 1$ response vector for the ith cluster or subject having conditional mean and variance-covariance matrix

$$E(\mathbf{y}_i|\mathbf{b}_i) = \boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i, \mathbf{x}_i, \mathbf{z}_i) = \boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i)$$

and

$$\operatorname{var}(\mathbf{y}_i|\mathbf{b}_i) = \boldsymbol{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i, \mathbf{x}_i, \mathbf{z}_i) = \boldsymbol{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i). \quad (1)$$

In general, the $p_i \times 1$ conditional mean vector, $\boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i)$, is a nonlinear function in both $\boldsymbol{\beta}$ and $\mathbf{b}_i$. The conditional variance-covariance matrix, $\boldsymbol{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i)$, is assumed to be positive-definite and possibly dependent on an unknown $u \times 1$ vector of variance-covariance parameters, $\boldsymbol{\alpha}$. As occurs with generalized linear and nonlinear mixed-effects models, $\boldsymbol{\Lambda}_i$ may also depend on the fixed- and random-effects parameters. It is tacitly assumed that $\boldsymbol{\mu}_i$ and $\boldsymbol{\Lambda}_i$ satisfy the necessary regularity conditions required in any nonlinear regression setting. In particular, it is assumed that $\boldsymbol{\tau}' = (\boldsymbol{\beta}' \ \boldsymbol{\alpha}') \in \boldsymbol{\Theta}$, where $\boldsymbol{\Theta}$ is a compact subspace of $\Re^{s+u}$ and the components of $\boldsymbol{\mu}_i$ and $\boldsymbol{\Lambda}_i$ are thrice-continuous differentiable functions in $\boldsymbol{\beta}$, $\boldsymbol{\alpha}$, and $\mathbf{b}_i$.

In addition, it is assumed that the third- and fourth-order moments of $\mathbf{y}_i|\mathbf{b}_i$ exist, which we define as follows. Let

$$\mathbf{s}_i(\boldsymbol{\beta}, \mathbf{b}_i) = \operatorname{vech}\left([\mathbf{y}_i - \boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i)][\mathbf{y}_i - \boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i)]'\right),$$
$$\boldsymbol{\sigma}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i) = \operatorname{vech}\left(\boldsymbol{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i)\right) = E[\mathbf{s}_i(\boldsymbol{\beta}, \mathbf{b}_i)|\mathbf{b}_i],$$

and

$$\mathbf{w}_i = \operatorname{vech}(\mathbf{y}_i\mathbf{y}_i') = \mathbf{B}_i \operatorname{vec}(\mathbf{y}_i\mathbf{y}_i'),$$

where $\mathbf{B}_i$ is a $(p_i(p_i+1)/2) \times p_i^2$ matrix of 0's and 1's such that $\operatorname{vech}(\mathbf{A}_i) = \mathbf{B}_i \operatorname{vec}(\mathbf{A}_i)$ for any $p_i \times p_i$ symmetric matrix $\mathbf{A}_i$. If we let

$$\mathbf{W}_i(\boldsymbol{\beta}, \mathbf{b}_i) = \begin{pmatrix} \mathbf{I}_{p_i} & \mathbf{0} \\ \mathbf{T}_i(\boldsymbol{\beta}, \mathbf{b}_i) & \mathbf{I}_{\frac{p_i(p_i+1)}{2}} \end{pmatrix},$$

where $\mathbf{T}_i(\boldsymbol{\beta}, \mathbf{b}_i) = \mathbf{B}_i\{[\boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i) \otimes \mathbf{I}_{p_i}] + [\mathbf{I}_{p_i} \otimes \boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i)]\}$, then the third- and fourth-order central moments are related to the raw moments as follows:

$$\begin{pmatrix} \boldsymbol{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i) & \boldsymbol{\Upsilon}_i^* \\ \boldsymbol{\Upsilon}_i^{*\prime} & \boldsymbol{\Gamma}_i^* \end{pmatrix} = \mathbf{W}_i(\boldsymbol{\beta}, \mathbf{b}_i) \begin{pmatrix} \boldsymbol{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i) & \boldsymbol{\Upsilon}_i \\ \boldsymbol{\Upsilon}_i' & \boldsymbol{\Gamma}_i \end{pmatrix} \times \mathbf{W}_i(\boldsymbol{\beta}, \mathbf{b}_i)',$$

where $\boldsymbol{\Upsilon}_i^* = \operatorname{cov}[(\mathbf{y}_i, \mathbf{w}_i)|\mathbf{b}_i]$, $\boldsymbol{\Upsilon}_i = \operatorname{cov}[(\mathbf{y}_i, \mathbf{s}_i)|\mathbf{b}_i]$, $\boldsymbol{\Gamma}_i^* = \operatorname{var}(\mathbf{w}_i|\mathbf{b}_i)$, and $\boldsymbol{\Gamma}_i = \operatorname{var}(\mathbf{s}_i|\mathbf{b}_i)$.

To apply our methods for both continuous and discrete data, we assume that the conditional probability density function of $\mathbf{y}_i$, say $\pi(\mathbf{y}_i|\mathbf{b}_i)$, belongs to the quadratic exponential family with probability density

$$\pi(\mathbf{y}_i|\mathbf{b}_i) = \exp\{\mathbf{y}_i'\boldsymbol{\lambda}_i + \mathbf{w}_i'\boldsymbol{\gamma}_i - \Delta_i(\boldsymbol{\lambda}_i, \boldsymbol{\gamma}_i) + \mathbf{c}_i(\mathbf{y}_i)\}, \quad (2)$$

where $\boldsymbol{\lambda}_i$ and $\boldsymbol{\gamma}_i$ are canonical parameters, which are expressed in terms of the conditional moments via $\boldsymbol{\lambda}_i'(\boldsymbol{\mu}_i, \boldsymbol{\sigma}_i) = (\lambda_{i1}, \ldots, \lambda_{ip_i})$, $\boldsymbol{\gamma}_i'(\boldsymbol{\mu}_i, \boldsymbol{\sigma}_i) = (\gamma_{i11}, \gamma_{i12}, \ldots, \gamma_{i1p_i}, \gamma_{i22}, \gamma_{i23}, \ldots)$, and $\Delta_i(\boldsymbol{\lambda}_i, \boldsymbol{\gamma}_i)$ is a normalization constant. This family of distributions was considered by Prentice and Zhao (1991) in the context of marginal models for multivariate discrete or continuous data. To complete the model specification, we assume that the $\mathbf{b}_i$ are independently and identically distributed multivariate normal random vectors with mean $\mathbf{0}$ and variance-covariance $\boldsymbol{\Psi}$. It should be noted that by assuming that the conditional distribution of $\mathbf{y}_i|\mathbf{b}_i$ comes from the quadratic exponential family, one can easily accommodate more general intracluster covariance structures within the generalized linear mixed-effects framework. This provides additional flexibility over the usual modeling assumption of conditional independence.

3. ESTIMATION

By evaluating the posterior means, $E(\mathbf{b}_i|\mathbf{y}_i)$, and variances, $E(\mathbf{b}_i\mathbf{b}_i'|\mathbf{y}_i)$, one can apply the EM algorithm to obtain maximum likelihood estimates of $\boldsymbol{\beta}$, $\boldsymbol{\alpha}$, and $\boldsymbol{\Psi}$ (e.g., Diggle et al. 1994; Fahrmeir and Tutz 1994). But implementation is complicated by the need for numerical integration to evaluate the posterior means and variances (e.g., McCulloch 1997). An alternative strategy, which we adopt here, is to approximate the conditional means and variances by the posterior modes and curvatures as was done by Laird and Louis (1982), Stiratelli et al. (1984), Lindstrom and Bates (1990), and Breslow and Clayton (1993). The algorithm thus consists of alternately maximizing the log-posterior to estimate $\boldsymbol{\beta}$, $\boldsymbol{\alpha}$, and $\mathbf{b}_i$ ($i = 1, \ldots, n$) given a current estimate of $\boldsymbol{\Psi}$, and then updating $\boldsymbol{\Psi}$ via an EM step using the posterior modes and curvatures (inverse information).

Specifically, for fixed $\boldsymbol{\theta} = \operatorname{vech}(\boldsymbol{\Psi})$, estimation of $\boldsymbol{\beta}$, $\boldsymbol{\alpha}$, and $\mathbf{b}_i$ can be carried out by maximizing the log-posterior

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(ignoring constants)

$$l(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}; \mathbf{y}) = \sum_{i=1}^{n} l_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i; \mathbf{y}_i) - \frac{1}{2}\sum_{i=1}^{n} \mathbf{b}_i'\boldsymbol{\Psi}^{-1}\mathbf{b}_i, \quad (3)$$

where $\mathbf{y}' = [\mathbf{y}_1' \cdots \mathbf{y}_n']$, $\mathbf{b}' = [\mathbf{b}_1' \cdots \mathbf{b}_n']$, and $l_i = \mathbf{y}_i'\boldsymbol{\lambda}_i + \mathbf{w}_i'\boldsymbol{\gamma}_i - \Delta_i(\boldsymbol{\lambda}_i, \boldsymbol{\gamma}_i) + \mathbf{c}_i(\mathbf{y}_i)$. By appealing directly to the results presented by Prentice and Zhao (1991), it can be shown that maximizing (3) using Fisher's method of scoring is equivalent to solving the following set of CGEE2:

$$U(\boldsymbol{\beta}, \boldsymbol{\alpha}) = \begin{pmatrix} \partial l/\partial \boldsymbol{\beta} \\ \partial l/\partial \boldsymbol{\alpha} \end{pmatrix} = \sum_{i=1}^{n} \mathbf{X}_i'\mathbf{V}_i^{-1}(\mathbf{Y}_i - \mathbf{F}_i) = \mathbf{0}, \quad (4)$$

$$U_1(\mathbf{b}_1) = \partial l/\partial \mathbf{b}_1 = \mathbf{Z}_1'\mathbf{V}_1^{-1}(\mathbf{Y}_1 - \mathbf{F}_1) - \boldsymbol{\Psi}^{-1}\mathbf{b}_1 = \mathbf{0}, \quad (5)$$

$$\vdots$$

$$U_n(\mathbf{b}_n) = \partial l/\partial \mathbf{b}_n = \mathbf{Z}_n'\mathbf{V}_n^{-1}(\mathbf{Y}_n - \mathbf{F}_n) - \boldsymbol{\Psi}^{-1}\mathbf{b}_n = \mathbf{0},$$

where

$$\mathbf{X}_i = \begin{pmatrix} \partial \boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i)/\partial \boldsymbol{\beta}' & \mathbf{0} \\ \partial \boldsymbol{\sigma}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i)/\partial \boldsymbol{\beta}' & \partial \boldsymbol{\sigma}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i)/\partial \boldsymbol{\alpha}' \end{pmatrix},$$

$$\mathbf{Z}_i = \begin{pmatrix} \partial \boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i)/\partial \mathbf{b}_i' \\ \partial \boldsymbol{\sigma}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i)/\partial \mathbf{b}_i' \end{pmatrix}, \qquad \mathbf{V}_i = \begin{pmatrix} \boldsymbol{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i) & \boldsymbol{\Upsilon}_i \\ \boldsymbol{\Upsilon}_i' & \boldsymbol{\Gamma}_i \end{pmatrix},$$

$$\mathbf{Y}_i = \begin{pmatrix} \mathbf{y}_i \\ \mathbf{s}_i \end{pmatrix}, \qquad \text{and} \qquad \mathbf{F}_i = \begin{pmatrix} \boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i) \\ \boldsymbol{\sigma}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i) \end{pmatrix}.$$

By writing $\mathbf{X}' = [\mathbf{X}_1' \cdots \mathbf{X}_n']$, $\mathbf{Z}' = \mathrm{diag}\{\mathbf{Z}_1', \ldots, \mathbf{Z}_n'\}$, $\mathbf{Y}' = [\mathbf{Y}_1' \cdots \mathbf{Y}_n']$, $\mathbf{F}' = [\mathbf{F}_1' \cdots \mathbf{F}_n']$, and $\mathbf{V} = \mathrm{diag}\{\mathbf{V}_1, \ldots, \mathbf{V}_n\}$, and letting $\boldsymbol{\tau}' = (\boldsymbol{\beta}'\boldsymbol{\alpha}')$, it can be shown that solving (4) and (5) via Fisher's method of scoring reduces to iteratively solving the familiar linear mixed-model equations

$$\begin{pmatrix} \mathbf{X}'\mathbf{V}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{V}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{V}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{V}^{-1}\mathbf{Z} + \widetilde{\boldsymbol{\Psi}}^{-1} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\tau}}^{u+1} \\ \hat{\mathbf{b}}^{u+1} \end{pmatrix} = \begin{pmatrix} \mathbf{X}'\mathbf{V}^{-1}\mathbf{Y}^* \\ \mathbf{Z}'\mathbf{V}^{-1}\mathbf{Y}^* \end{pmatrix}, \quad (6)$$

where $\widetilde{\boldsymbol{\Psi}} = \mathrm{diag}\{\boldsymbol{\Psi}, \ldots, \boldsymbol{\Psi}\}$, $\mathbf{Y}^* = \mathbf{Y} - \mathbf{F} + \mathbf{X}\hat{\boldsymbol{\tau}}^u + \mathbf{Z}\hat{\mathbf{b}}^u$ and where $\mathbf{F}$, $\mathbf{X}$, $\mathbf{Z}$, and $\mathbf{V}$ are all evaluated at $(\hat{\boldsymbol{\tau}}^u, \hat{\mathbf{b}}^u)$. Solving (6) yields the one-step Gauss–Newton estimates

$$\hat{\boldsymbol{\tau}}^{u+1} = \left(\sum_{i=1}^{n} \mathbf{X}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{X}_i\right)^{-1} \sum_{i=1}^{n} \mathbf{X}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{Y}_i^*$$

and

$$\hat{\mathbf{b}}_i^{u+1} = \boldsymbol{\Psi}\mathbf{Z}_i'\boldsymbol{\Sigma}_i^{-1}\left(\mathbf{Y}_i^* - \mathbf{X}_i\hat{\boldsymbol{\tau}}^{u+1}\right) \quad (i = 1, \ldots, n), \quad (7)$$

where $\boldsymbol{\Sigma}_i = \mathbf{Z}_i\boldsymbol{\Psi}\mathbf{Z}_i' + \mathbf{V}_i$. By fully iterating (7), we obtain CGEE2 estimates $\hat{\boldsymbol{\tau}}' = \hat{\boldsymbol{\tau}}(\boldsymbol{\theta})' = (\hat{\boldsymbol{\beta}}(\boldsymbol{\theta})'\hat{\boldsymbol{\alpha}}(\boldsymbol{\theta})')$ and $\hat{\mathbf{b}} = \hat{\mathbf{b}}(\boldsymbol{\theta})$, each of which depends on the current value of $\boldsymbol{\theta} = \mathrm{vech}(\boldsymbol{\Psi})$.

To complete one cycle of the algorithm, we estimate $\boldsymbol{\Psi}$ using the Gaussian posterior approximation of Laird and Louis (1982). Specifically, if $\widehat{\boldsymbol{\Psi}}^w$ is the current estimate of $\boldsymbol{\Psi}$ and $\hat{\boldsymbol{\tau}}(\hat{\boldsymbol{\theta}}^w)$ and $\hat{\mathbf{b}}_i(\hat{\boldsymbol{\theta}}^w)$ are the current CGEE2 estimates of $\boldsymbol{\tau}$ and $\mathbf{b}_i$ given $\hat{\boldsymbol{\theta}}^w = \mathrm{vech}(\widehat{\boldsymbol{\Psi}}^w)$, then

$$\widehat{\boldsymbol{\Psi}}^{w+1} = \sum_{i=1}^{n} \left(\boldsymbol{\Omega}(\hat{\mathbf{b}}_i(\hat{\boldsymbol{\theta}}^w)) + \hat{\mathbf{b}}_i(\hat{\boldsymbol{\theta}}^w)\hat{\mathbf{b}}_i(\hat{\boldsymbol{\theta}}^w)'\right)/n, \quad (8)$$

where we have replaced the posterior mean in an EM-type algorithm by the posterior mode, $\hat{\mathbf{b}}_i(\boldsymbol{\theta})$, and the posterior variance by the posterior curvature, $\boldsymbol{\Omega}(\hat{\mathbf{b}}_i(\boldsymbol{\theta}))$, the latter of which can be computed from the inverse of the Fisher information matrix in (6). In particular, given $\boldsymbol{\theta}$, the posterior curvature for $\hat{\boldsymbol{\tau}}(\boldsymbol{\theta})' = (\hat{\boldsymbol{\beta}}(\boldsymbol{\theta})'\hat{\boldsymbol{\alpha}}(\boldsymbol{\theta})')$ is

$$\boldsymbol{\Omega}(\hat{\boldsymbol{\tau}}(\boldsymbol{\theta})) = \left(\sum_{i=1}^{n} \mathbf{X}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{X}_i\right)^{-1}, \quad (9)$$

whereas the posterior curvature for $\hat{\mathbf{b}}_i(\boldsymbol{\theta})$ is

$$\boldsymbol{\Omega}(\hat{\mathbf{b}}_i(\boldsymbol{\theta})) = \boldsymbol{\Psi} - \boldsymbol{\Psi}\mathbf{Z}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{Z}_i\boldsymbol{\Psi} + \boldsymbol{\Psi}\mathbf{Z}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{X}_i\boldsymbol{\Omega}(\hat{\boldsymbol{\tau}}(\boldsymbol{\theta}))\mathbf{X}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{Z}_i\boldsymbol{\Psi}. \quad (10)$$

By evaluating (10) at $\widehat{\boldsymbol{\Psi}}^w$ and substituting into (8), we obtain an updated estimate, $\widehat{\boldsymbol{\Psi}}^{w+1}$, which is identical in form to the EM-based restricted maximum likelihood (REML) estimator for linear mixed-effects models (e.g., Laird, Lange, and Stram 1987). Alternatively, by treating $\boldsymbol{\beta}$ and $\boldsymbol{\alpha}$ as though they were fixed, we can compute a pseudo-ML estimator of $\boldsymbol{\Psi}$ by replacing $\boldsymbol{\Omega}(\hat{\mathbf{b}}_i(\boldsymbol{\theta}))$ with

$$\boldsymbol{\Omega}(\hat{\mathbf{b}}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \boldsymbol{\theta})) = \boldsymbol{\Psi} - \boldsymbol{\Psi}\mathbf{Z}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{Z}_i\boldsymbol{\Psi}. \quad (11)$$

More generally, one can set $\boldsymbol{\Psi}(\boldsymbol{\theta})$ to be a $v \times v$ positive-definite covariance matrix that depends on a vector of covariance parameters, $\boldsymbol{\theta}$, and simply work with the usual normal-theory profile likelihood function to obtain an ML or an REML estimate of $\boldsymbol{\Psi}(\boldsymbol{\theta})$ given current estimates of $\boldsymbol{\beta}$, $\boldsymbol{\alpha}$, and $\mathbf{b}$ (e.g., Breslow and Clayton 1993).

4. CONDITIONAL VARIANCE-COVARIANCE STRUCTURES

Although estimation based on CGEE2 is conceptually quite simple, specification of the third- and fourth-order central moments that define $\mathbf{V}_i$ can be quite problematic, as indicated by Prentice and Zhao (1991). In the general setting, one would need an iterative calculation to compute the canonical parameters $\boldsymbol{\lambda}_i$ and $\boldsymbol{\gamma}_i$. From these, one would then calculate the third- and fourth-order moments, $\boldsymbol{\Upsilon}_i$ and $\boldsymbol{\Gamma}_i$, directly by summation or integration.

In some cases, the third- and fourth-order moments may depend on an additional set of parameters, say $\boldsymbol{\eta}$, in which case we write $\boldsymbol{\Upsilon}_i = \boldsymbol{\Upsilon}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \boldsymbol{\eta})$ and $\boldsymbol{\Gamma}_i = \boldsymbol{\Gamma}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \boldsymbol{\eta})$. When this occurs, CGEE2 requires either that $\boldsymbol{\eta}$ be fixed and known or that it can be replaced by some consistent estimate. Alternatively, one can adopt the approach taken by Prentice and Zhao for the marginal setting by specifying a conditional "working" variance matrix $\mathbf{V}_i$ through specification of $\boldsymbol{\Upsilon}_i$ and $\boldsymbol{\Gamma}_i$ as functions of $\boldsymbol{\mu}_i$ and $\boldsymbol{\sigma}_i$. In this section we present several conditional variance-covariance structures that may be suitable for use with generalized linear and nonlinear mixed-effects models.

In many random-effects settings, a commonly used assumption is that of conditional independence; namely, that conditional on the random effects $\mathbf{b}_i$, the components y_{ik} $(k = 1, \ldots, p_i)$ are independently distributed with means μ_{ik} and

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variances σ_{ikk}. In this case the structure of $\mathbf{\Lambda}_i$, $\mathbf{\Upsilon}_i$, and $\mathbf{\Gamma}_i$ are easily obtained based on the underlying distribution assumptions. Moreover, one can further simplify the structure of $\mathbf{V}_i$ by noting that under conditional independence, the off-diagonal elements of the third-order central moments will be 0 except for the terms $(y_{ik}-\mu_{ik})^3$. Thus one might consider specifying a conditional "working" independence structure by simply setting $\mathbf{\Upsilon}_i = \mathbf{0}$. This makes the numerical computations relatively easy, because the matrices, $\mathbf{V}_i$, will now be block diagonal with both $\mathbf{\Lambda}_i$ and $\mathbf{\Gamma}_i$ themselves being diagonal.

One can expand on the conditional independence or conditional "working" independence structures by allowing for overdispersion parameters in the specification of the conditional variances. For example, when modeling count data via a mixed-effects Poisson model, one can allow for intracluster overdispersion by setting $\sigma_{ikk} = \alpha\mu_{ik}$, where $\alpha > 0$ is the overdispersion parameter. More complicated models with overdispersion can also be accommodated. For example, by setting $\alpha_i = \exp(\boldsymbol{\alpha}'\mathbf{x}_i^*)$, where $\mathbf{x}_i^*$ is some subset of the covariates, $\mathbf{x}_i$, one can model Poisson overdispersion by setting $\sigma_{ikk} = \alpha_i\mu_{ik}$ and apply CGEE2 to jointly estimate the mean and dispersion parameters $\boldsymbol{\beta}$ and $\boldsymbol{\alpha}$. Moreover, CGEE2 allows for one to estimate parameters from conditional distributions like the negative binomial, where there may be additional nonlinear parameters in the variance function that are not covered in the usual generalized linear model setting (e.g., McCullagh and Nelder 1989, chap. 11).

To accommodate within-cluster correlation, one can combine the conditional independence structure with a "working" correlation matrix in much the same way that Liang and Zeger (1986) do for marginal models. For example, let $\boldsymbol{\alpha}_1$ denote a set of variance parameters assuming conditional independence (e.g., a set of dispersion parameters from a normal distribution) and let $\mathbf{R}_i(\boldsymbol{\alpha}_2)$ represent a "working" correlation matrix with correlation parameter vector $\boldsymbol{\alpha}_2$. By combining these two structures, one can form the conditional "working" variance-covariance matrix

$$\mathbf{V}_i = \begin{pmatrix} \mathbf{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}_1, \mathbf{b}_i) & \mathbf{\Upsilon}_i \\ \mathbf{\Upsilon}_i' & \mathbf{\Gamma}_i \end{pmatrix}^{1/2} \times \begin{pmatrix} \mathbf{R}_i(\boldsymbol{\alpha}_2) & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_i \end{pmatrix} \begin{pmatrix} \mathbf{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}_1, \mathbf{b}_i) & \mathbf{\Upsilon}_i \\ \mathbf{\Upsilon}_i' & \mathbf{\Gamma}_i \end{pmatrix}^{1/2}, \quad (12)$$

where $\mathbf{\Lambda}_i$, $\mathbf{\Upsilon}_i$, and $\mathbf{\Gamma}_i$ are as defined earlier. A slightly simpler version of $\mathbf{V}_i$ can be obtained by replacing the conditional independence structure with the conditional "working" independence structure (i.e., setting $\mathbf{\Upsilon}_i = \mathbf{0}$). One could also add to the structure of $\mathbf{V}_i$ by including, for example, parameters that describe possible overdispersion. One might even consider extending the "working" correlation to include the fourth-order moments by replacing the identity matrix in the foregoing expression with an "extended working" correlation matrix. In any case, it should be stressed that by specifying a "working" correlation matrix, we are tacitly assuming that the "working" covariance matrix, $\mathbf{V}_i$, is the "true" covariance matrix. This assumption is needed if we are to avoid the "uncertainty of definition" issues that occur with the use of "working" correlation structures (e.g., Crowder 1995).

Under general Gaussian assumptions, the matrix $\mathbf{V}_i$ is given by

$$\mathbf{V}_i = \begin{pmatrix} \mathbf{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i) & \mathbf{0} \\ \mathbf{0} & \mathbf{\Gamma}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i) \end{pmatrix}, \quad (13)$$

where $\mathbf{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i)$ is any positive-definite covariance matrix (not necessarily diagonal) and

$$\mathbf{\Gamma}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i) = \left\{\mathbf{B}_i[\mathbf{I}_{p_i^2} + \mathbf{I}_{(p_i, p_i)}](\mathbf{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i) \otimes \mathbf{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i))\mathbf{B}_i'\right\}$$

is the variance-covariance structure of $\mathbf{s}_i(\boldsymbol{\beta}, \mathbf{b}_i)$. Here, $\mathbf{I}_{(p_i, p_i)}$ is a $p_i^2 \times p_i^2$ permuted identity matrix (e.g., Henderson and Searle 1981) and $\mathbf{B}_i$ is as defined earlier. As noted later, this structure also arises when CGEE2 is used to minimize a penalized least squares objective function.

5. CONNECTIONS WITH PENALIZED QUASI-LIKELIHOOD AND PENALIZED EXTENDED LEAST SQUARES

The CGEE2 estimation procedure is formally derived assuming that the conditional distribution of $\mathbf{y}_i|\mathbf{b}_i$ comes from a quadratic exponential family. If the conditional distribution is from the linear exponential family, then CGEE2 yields estimates identical to the penalized quasi-likelihood (PQL) estimates described by Breslow and Clayton (1993). This follows from the fact that under the linear exponential family, $\mathbf{w}_i'\boldsymbol{\gamma}_i = 0$, and hence the second-order moments will contain no additional information about $\boldsymbol{\beta}$ or $\mathbf{b}_i$ not already contained in $\mathbf{y}_i'\boldsymbol{\lambda}_i$. This has important implications in that one can apply the asymptotic properties of the CGEE2 estimator to the PQL estimator whenever the conditional distribution is from the linear (regular) exponential family.

A second feature of CGEE2 is that it can be used to obtain penalized least mean distance estimators of $\boldsymbol{\beta}$, $\boldsymbol{\alpha}$, and $\mathbf{b}$ without necessarily requiring one to specify the conditional distribution of $\mathbf{y}_i|\mathbf{b}_i$. For instance, suppose that one were to model possible serial correlation among a set of repeated measurements using a conditional "working" correlation structure. Then, depending on the data, it may or may not be reasonable to assume that the first two moments correspond to a conditional distribution from the quadratic exponential family. In this case, rather than seek a likelihood-based analysis, one might consider some sort of penalized least squares analysis, taking into account a desire to model intracluster variance heterogeneity and a desire to obtain smoothed shrinkage-type estimates of the cluster-specific random effects.

One possibility for such an approach would be to jointly estimate $\boldsymbol{\beta}$, $\boldsymbol{\alpha}$, and $\mathbf{b}$ by minimizing the penalized generalized least squares (PGLS) objective function

$$Q_{\text{PGLS}}(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}|\boldsymbol{\Psi}) = \sum_{i=1}^{n}\left\{(\mathbf{y}_i - \boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i))'\mathbf{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i)^{-1} \times (\mathbf{y}_i - \boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i))\right\} + \sum_{i=1}^{n}\mathbf{b}_i'\boldsymbol{\Psi}^{-1}\mathbf{b}_i. \quad (14)$$

The problem with minimizing (14) is that even when $\boldsymbol{\alpha}$ and $\mathbf{b}$ are known, the corresponding PGLS estimating equations for

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$\boldsymbol{\beta}$ will be biased (e.g., Vonesh, Wang, and Majumdar 2001). Using arguments similar to those of Vonesh et al. (2001), one can eliminate the bias by applying a simple method-of-moments correction factor. By then extending this correction factor to include $\boldsymbol{\alpha}$, one attains a set of penalized extended least squares (PELS) estimating equations that are conditionally unbiased for $\boldsymbol{\beta}$ and $\boldsymbol{\alpha}$. It can be shown that the estimating equations obtained by minimizing the PELS objective function,

$$Q_{\mathrm{PELS}}(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b} \mid \boldsymbol{\Psi}) = \sum_{i=1}^{n} \{ (\mathbf{y}_i - \boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i))' \boldsymbol{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i))^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_i(\boldsymbol{\beta}, \mathbf{b}_i)) + \log |\boldsymbol{\Lambda}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i)| \} + \sum_{i=1}^{n} \mathbf{b}_i' \boldsymbol{\Psi}^{-1} \mathbf{b}_i, \quad (15)$$

coincide with the CGEE2 estimating equations (4) and (5) provided that $\mathbf{V}_i$ has the Gaussian "working" structure (13). This follows from the fact that under normality, minimizing the PELS objective function (15) is equivalent to maximizing the log-posterior (3). However, as with extended least squares for marginal models (Vonesh et al. 2001), the PELS objective function may be derived independent of any distributional assumptions. Thus, by specifying $\mathbf{V}_i$ to have the "working" Gaussian structure (13), one can apply the CGEE2 algorithm to obtain a PELS estimator of $\boldsymbol{\tau}' = (\boldsymbol{\beta}' \ \boldsymbol{\alpha}')$ independent of any distributional assumptions. The advantage of PELS is that one need specify only the first two conditional moments.

6. ASYMPTOTIC PROPERTIES

Let $\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}} = (\hat{\boldsymbol{\beta}}_{\mathrm{CGEE2}}, \hat{\boldsymbol{\alpha}}_{\mathrm{CGEE2}})$ be the CGEE2 estimator of $\boldsymbol{\tau}$ satisfying $U(\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}}) = \mathbf{0}$, where $U(\boldsymbol{\tau})$ is the set of CGEE2 (4). It is shown in the Appendix that under fairly mild regularity conditions, $\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}}$ satisfies the following properties:

$$\begin{aligned} \hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}} &= \hat{\boldsymbol{\tau}}_{\mathrm{MLE}} + O_p\{ \min(p_i)^{-1/2} \}, \\ &= \boldsymbol{\tau} + O_p\{ \max[n^{-1/2}, \min(p_i)^{-1/2}] \}, \end{aligned} \quad (16)$$

where $\hat{\boldsymbol{\tau}}_{\mathrm{MLE}}$ denotes the unconditional ML estimator. Thus the CGEE2 estimator is not only consistent but also asymptotically equivalent to the ML estimator. The order of convergence is shown to depend on both the cluster size (or number of observations per subject) and the number of clusters (subjects). This result, although weaker, is similar to that obtained by Vonesh (1996) for the closely related Laplace-based ML estimator of a Gaussian nonlinear mixed-effects model. There the order of convergence is shown to be $O_p\{\max[n^{-1/2}, \min(p_i)^{-1}]\}$, assuming that the conditional covariance structure is independent of the random effects. This stronger order of convergence is due to the fact that the Laplace-based ML method uses Laplace's approximation to directly approximate the marginal likelihood, whereas CGEE2 uses Laplace's approximation to approximate the conditional (on $\boldsymbol{\Psi}$) ML score equations for $\boldsymbol{\tau}$ (see the Appendix). In the setting described by Vonesh (1996), the former method approximates the marginal likelihood, whereas the latter corresponds to approximating a modified profile likelihood (e.g., Wolfinger 1993). Our results also agree with those obtained by Demidenko (1997), who showed that under a Gaussian nonlinear random-effects setting, the two-stage, Lindstrom and Bates, and ML estimators will all be asymptotically equivalent provided that both n and $\min(p_i) \to \infty$. However, our results apply to a broader class of mixed-effects models that includes generalized linear and nonlinear mixed models. In fact, our results also apply to the PQL estimator of Breslow and Clayton (1993), provided that the conditional distribution is from the linear exponential family.

In terms of inference, the CGEE2 estimator has a limiting normal distribution with $\sqrt{n}(\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}} - \boldsymbol{\tau}) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Omega}(\boldsymbol{\tau}))$. The asymptotic covariance matrix, $\boldsymbol{\Omega}(\boldsymbol{\tau})$, can be consistently estimated by the inverse Fisher information matrix (9), provided that the conditional moments have not been misspecified. As an alternative to (9), one may consider using the robust "sandwich" estimator

$$\boldsymbol{\Omega}_{\mathrm{R}}(\hat{\boldsymbol{\tau}}(\boldsymbol{\theta})) = \boldsymbol{\Omega}(\hat{\boldsymbol{\tau}}(\boldsymbol{\theta})) \left(\sum_{i=1}^{n} \mathbf{X}_i' \boldsymbol{\Sigma}_i^{-1} \mathbf{e}_i \mathbf{e}_i' \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i \right) \boldsymbol{\Omega}(\hat{\boldsymbol{\tau}}(\boldsymbol{\theta})), \quad (17)$$

where $\mathbf{e}_i = \mathbf{Y}_i^* - \mathbf{X}_i \hat{\boldsymbol{\tau}}$ is evaluated at the final estimate of $\boldsymbol{\tau}$. Finally, because $\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}}$ is asymptotically equivalent to the ML estimator, it will be asymptotically efficient as $\min(p_i) \to \infty$.

7. SIMULATIONS

As noted previously, CGEE2 and PQL yield equivalent estimators provided that the conditional distribution is from the linear exponential family. Consequently, for highly discrete data, such as occur with binary outcomes, one might anticipate that CGEE2 will perform less satisfactory than more "exact" ML methods using numerical integration techniques (e.g., Breslow and Clayton 1993; Rodriguez and Goldman 1995; Lin and Breslow 1996; McCulloch 1997; Neuhaus and Segal 1997). However, with increasing cluster size, the CGEE2 or PQL estimator can be expected to perform at or near the level of the computationally more intensive ML estimator (e.g., Ten Have and Localio 1999). To see this, we compared, via simulation, the consistency and asymptotic efficiency of the CGEE2 estimator with estimates obtained as solutions to a set of CGEE1 and with estimates obtained via numerical integration (ML). In particular, the CGEE1 estimates are those obtained using the PQL algorithm of Breslow and Clayton (1993) or, equivalently, the nonlinear mixed effects (NLME) algorithm of Lindstrom and Bates (1990) and Pinheiro and Bates (2000). We compared the three estimators using simulated data from a Gaussian-based nonlinear mixed-effects model. We also evaluated CGEE2 using simulated data from a discrete multivariate Poisson mixed-effects model, as well as simulated data from a binary mixed-effects logistic regression model.

For the Gaussian-based nonlinear mixed-effects model, we generated data assuming the following negative exponential decay model:

$$y_{ij} = e^{-(\beta - \sqrt{\psi} b_i) t_{ij}} (1 + \epsilon_{ij}), \qquad i = 1, \ldots, 25; \quad j = 1, \ldots, 6, \quad (18)$$

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where $\beta = 2.5$, $t_{ij} = \{.1, .2, .3, .4, .7, 1.0\}$, $b_i \sim N(0, 1)$, and $\boldsymbol{\epsilon}_i = (\epsilon_{i1}\epsilon_{i2}\cdots\epsilon_{i6})' \sim N(\mathbf{0}, \sigma^2\boldsymbol{\Lambda}(\boldsymbol{\alpha}_1))$. We varied the correlation matrix, $\boldsymbol{\Lambda}(\boldsymbol{\alpha}_1)$, by assuming one of three structures: independence ($\boldsymbol{\alpha}_1 = 0$), compound symmetry ($\boldsymbol{\alpha}_1 = \rho = .50$), or first-order autoregressive ($\boldsymbol{\alpha}_1 = \rho = .50$). In all three cases, the conditional variance depends on the fixed- and random-effects parameters, β and b_i.

We created 200 sample datasets based on randomly generated values of b_i and ϵ_{ij}. To determine the impact of the size of the variance on each estimator, we compared results across two sets of values of (σ^2, ψ). The first set corresponds to small-to-moderate variation ($\sigma^2 = .0625, \psi = .25$), and the second set corresponds to moderate-to-large variation ($\sigma^2 = .49, \psi = 1.0$). ML estimates for the conditional independence and conditional compound symmetry structures were obtained using the SAS version 8.0 procedure NLMIXED. The CGEE1 estimates were obtained using the SAS macro NLINMIX (Littell, Milliken, Stroup, and Wolfinger 1996), and the CGEE2 estimates were obtained using a modification of NLINMIX available from the authors.

Results of this simulation are shown in Table 1. Under conditional independence with small-to-moderate variation, all three methods yield comparable results. However, for larger variation, a significant reduction in mean squared error (MSE)

Table 1. Simulation Results for the Gaussian Nonlinear Mixed-Effects Model (18)

		Variance parameters					
		$\sigma^2 = .0625, \psi = .25$ Method			$\sigma^2 = .49, \psi = 1.0$ Method		
Parameter	*Statistic*	*CGEE1*	*CGEE2*	*MLE*	*CGEE1*	*CGEE2*	*MLE*
Conditional independence							
	m	200	200	195	109	181	198
$\beta = 2.5$	Mean	2.5131	2.5225	2.4971	2.5488	2.6114	2.5110
	SD	.1049	.1045	.1028	.2642	.2392	.2373
	MSE	.0111	.0114	.0105	.0715	.0693	.0562
σ^2	Mean	.0644	.0542	.0627	.6617	.4615	.4954
	SD	.0091	.0077	.0087	.3011	.0675	.0676
	MSE	.0001	.0001	.0001	.1193	.0053	.0046
ψ	Mean	.2596	.2637	.2444	.9112	.9955	.9226
	SD	.0828	.0828	.0797	.3553	.3086	.2913
	MSE	.0069	.0070	.0063	.1330	.0947	.0904
b_i	Mean bias[a]	.0009	.0009	.0320	.0094	−.0032	.1001
	MSE[b]	.0417	.0392	.0399	.3330	.2019	.2116
Conditional compound symmetry							
	m	200	104	181	49	138	146
$\beta = 2.5$	Mean	2.5216	2.5346	2.5008	2.5515	2.6223	2.4780
	Observed SD	.1089	.1076	.1084	.2819	.2457	.2268
	MSE	.0123	.0127	.0117	.0805	.0749	.0516
σ^2	Mean	.0624	.0558	.0618	.5613	.4701	.4975
	Observed SD	.0128	.0123	.0122	.2138	.1078	.0923
	MSE	.0002	.0002	.0001	.0499	.0119	.0085
ψ	Mean	.2632	.2547	.2441	.9141	1.0256	.8977
	Observed SD	.0866	.0812	.0824	.4466	.3257	.3031
	MSE	.0076	.0065	.0068	.2028	.1060	.1017
$\rho = .50$	Mean	.4557	.4912	.4686	.3508	.4768	.4903
	Observed SD	.1239	.1224	.1204	.1557	.1350	.1001
	MSE	.0172	.0149	.0154	.0460	.0186	.0101
b_i	Mean bias[a]	.0009	−.0009	.0415	−.0020	−.0050	.1429
	MSE[b]	.0523	.0476	.0502	.4163	.2422	.2458
Conditional first-order autoregressive							
	m	200	178		47	168	
$\beta = 2.5$	Mean	2.5249	2.5310		2.5648	2.6492	
	Observed SD	.1120	.1197		.2770	.2413	
	MSE	.0131	.0152		.0793	.0801	
σ^2	Mean	.0642	.0897		.7205	.4958	
	Observed SD	.0121	.1413		.5339	.3440	
	MSE	.0001	.0206		.3217	.1177	
ψ	Mean	.2649	.3005		1.0483	1.0365	
	Observed SD	.0930	.1175		.5230	.3526	
	MSE	.0088	.0163		.2700	.1249	
$\rho = .50$	Mean	.4747	.3662		.3207	.4453	
	Observed SD	.0984	.2166		.2205	.1333	
	MSE	.0103	.0645		.0797	.0207	
b_i	Mean bias[a]	.0009	.0025		−.0027	−.0037	
	MSE[b]	.0642	.0851		.4717	.2620	

[a] $M^{-1}\sum(\hat{b}_i - b_i)$, where $M = m \times 25$ and m =# converged.
[b] $M^{-1}\sum(\hat{b}_i - b_i)^2$.

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was achieved with ML and CGEE2 compared to CGEE1 for both the fixed- and random-effects parameters. Similar results were also achieved under conditional compound symmetry, with ML and CGEE2 generally outperforming CGEE1 for larger intra- and interindividual variation. Under the conditional AR(1) structure, the results were mixed, with CGEE1 outperforming CGEE2 for small-to-moderate variation and CGEE2 outperforming CGEE1 for moderate-to-large variation. In the two cases where direct comparisons were made, the performance of CGEE2 was near that of ML. Finally, the ML and CGEE2 algorithms proved to be more stable than CGEE1, in that both showed better convergence behavior with increasing variation. Although this may be expected for a sophisticated ML procedure like NLMIXED, the better convergence behavior of CGEE2 compared to CGEE1 for large inter- and intrasubject variation probably reflects its ability to better approximate the ML score equations when the conditional distribution is from the quadratic exponential family (see the Appendix).

In our second set of simulations, we extend the simulation study described by Vonesh et al. (2001) to a mixed-effects setting. Specifically, Poisson count data were generated according to the following two-stage model:

$$y_{ij}|b_i = u_i + z_{ij} \quad (i = 1, \ldots, 30;\ j = 1, 2, \ldots, 8), \tag{19}$$

where the b_i are iid $N(0, \Psi)$, the u_i are iid Poisson random variables with mean $\mu = e^{\alpha}$, and the z_{ij} are conditionally (on b_i) independent Poisson random variables with log mean counts, $\log(\mu_{ij}) = \beta_0 + (\beta_1 + b_i)x_{ij} + \beta_2 G_i + \beta_3 x_{ij} \times G_i$. The b_i, u_i, and z_{ij} were generated independent of each other. Values of the covariates are $x_{ij} = \{.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$ for all $i = 1, \ldots, 30$, $G_i = 1$ for $i \leq 15$, and $G_i = 0$ for $i > 15$. Two sets of parameters were used, one set corresponding to data with small-to-moderate Poisson counts (i.e., $\alpha = -.50$, $\beta_0 = -2.5, \beta_1 = 1.0, \beta_2 = -1.0, \beta_3 = 0$) and one set corresponding to data with moderate-to-large Poisson counts (i.e., $\alpha = -.25, \beta_0 = -.5, \beta_1 = 1.0, \beta_2 = -1.0, \beta_3 = 0$).

Letting $\boldsymbol{\beta}' = (\beta_0, \beta_1, \beta_2, \beta_3, \alpha) \in \Re^5$, the conditional distribution of $y|b$ corresponds to a multivariate Poisson model with

$$\begin{aligned} E(y_{ij}|b_i) &= \mu_{ij}(\boldsymbol{\beta}, b_i) = e^{\alpha} + e^{\beta_0 + (\beta_1 + b_i)x_{ij} + \beta_2 G_i + \beta_3 x_{ij} \times G_i}, \\ \mathrm{var}(y_{ij}|b_i) &= \mu_{ij}(\boldsymbol{\beta}, b_i),\ \ \mathrm{cov}(y_{ij}, y_{ij'}|b_i) = e^{\alpha} \quad (j \neq j'). \end{aligned} \tag{20}$$

Although α appears in the conditional mean, it may be thought of primarily as a conditional covariance parameter in that it exclusively describes the conditional covariance between any two repeated counts. This conditional covariance structure parallels compound symmetry or the exchangeable covariance structure of a linear random-effects model, but it differs in that the variances and correlations vary with the means. The third- and fourth-order conditional moments are readily calculated based on the third- and fourth-order moments of independent Poisson random variables as $\Upsilon_{i(jk,l)} = \mathrm{cov}(z_{ij}z_{ik}, z_{il}) + \mathrm{var}(u_i)$ and $\Gamma_{i(jk,lm)} = \mathrm{cov}(z_{ij}z_{ik}, z_{il}z_{im}) + \mathrm{var}(u_i^2) + \mathrm{var}(u_i) \times \mathrm{cov}[z_{ij} + z_{ik}, z_{il} + z_{im}]$.

We carried out simulation results for this model using CGEE2. We also tried to fit the data with CGEE1 assuming conditional independence, that is, assuming $\mathrm{cov}(y_{ij}, y_{ij'}|b_i) = 0$. However, all attempts to fit the data to this "working" version of model (20) via CGEE1 failed, thus illustrating at least one setting in which CGEE1 fails even under the simplest of working assumptions. We also performed a sensitivity analysis by fitting the data generated under model (20) to the following closely aligned Poisson mixed-effects model with conditional independence:

$$\begin{aligned} E(y_{ij}|\mathbf{b}_i) &= \mu^*_{ij}(\boldsymbol{\beta}, \mathbf{b}_i) = e^{(\beta_0 + b_{0i}) + (\beta_1 + b_{1i})x_{ij} + \beta_2 G_i + \beta_3 x_{ij} \times G_i}, \\ \mathrm{var}(y_{ij}|\mathbf{b}_i) &= \mu^*_{ij}(\boldsymbol{\beta}, \mathbf{b}_i),\ \ \mathrm{cov}(y_{ij}, y_{ij'}|\mathbf{b}_i) = 0 \quad (j \neq j'). \end{aligned} \tag{21}$$

By fitting the data to model (21) rather than to model (20), we are incorrectly assuming that the random intercept term enters the model multiplicatively rather than additively, and that correlation is induced strictly through the random effects, $\mathbf{b}_i$, which we assumed to be bivariate normal.

Results of this simulation are shown in Table 2. Although models (20) and (21) both share a common set of fixed-effects parameters, namely $\beta_0, \beta_1, \beta_2$, and β_3, as well as a common random-effects parameter, b_{1i}, they differ in terms of how the overall intercept and the conditional covariance structure are modeled. Consequently, it is not surprising that CGEE2 estimates computed under the correctly specified model (20) are unbiased and more efficient than CGEE2 (≡CGEE1≡PQL) estimates computed under the misspecifications of model (21). Although not considered here, the reverse may hold for data generated under model (21) but analyzed assuming model (20). Despite its limited scope, this simulation nevertheless illustrates a setting where the standard assumption of conditional independence in the generalized linear mixed-effects setting is violated and where one may improve on current first-order methods like PQL by directly including the conditional second-order moments in the set of underlying estimating equations.

Finally, we replicated part of the simulation study described by Breslow and Clayton (1993) to evaluate the performance of CGEE2 (≡PQL) when the cluster size for binary outcomes increases. A total of 200 datasets were generated, with each dataset consisting of $n = 100$ clusters and $p_i = 28$ distinct binary observations per cluster. Conditionally, independent binary observations were generated according to the same mixed-effects logistic regression model described by Breslow and Clayton (1993) but with the within-cluster covariate, t_{ij}, taking distinct values at $t_{ij} = v_j - 4$, where v_j is varied systematically from 1.00 to 7.75 in .25 increments. Specifically, data were generated assuming $E(y_{ij}|b_i) = \pi_{ij}$, $\mathrm{var}(y_{ij}|b_i) = \pi_{ij}(1 - \pi_{ij})$, and $\log(\pi_{ij}/(1 - \pi_{ij})) = (\beta_0 + b_i) + \beta_1 t_{ij} + \beta_2 x_i + \beta_3 t_{ij} x_i$, where x_i is a between-cluster covariate taking values $x_i = 1$ for one half of the sample and $x_i = 0$ for the other half. As with Breslow and Clayton, the regression coefficients were fixed at $\boldsymbol{\beta}' = (-2.5, 1.0, -1.0, -.5)$ and the random intercepts, b_i, were generated independently as iid $N(0, \Psi)$ with $\Psi = 1$. The mean and standard deviation of the parameters estimates from our simulation were $\hat{\beta}_0 = -2.38(\pm.17)$, $\hat{\beta}_1 = .96(\pm.06)$, $\hat{\beta}_2 = -.94(\pm.27)$, $\hat{\beta}_3 = -.47(\pm.08)$, and $\widehat{\Psi} = .85(\pm.20)$. As one might have anticipated, these results are virtually identical with those presented by Breslow and Clayton (1993) for the conditionally independent binomial setting in which binomial observations with denominator, $m = 4$, were generated at each

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Table 2. Simulation Results for the Multivariate Poisson Mixed-Effects Model (20)

		Data generated under model (20)			
		Small counts ($\alpha = -.50, \beta_0 = -2.5$)		Large counts ($\alpha = -.25, \beta_0 = -.5$)	
Parameter	Statistic	Model (20) CGEE2	Model (21) CGEE2*	Model (20) CGEE2	Model (21) CGEE2*
	m	187	200	142	200
α	Mean	−.5758	–	−.3948	–
	SD	.2534	–	.3948	–
	MSE	.0701	–	.1796	–
β_0	Mean	−2.5377	−1.2901	−.4804	−.1634
	SD	.5026	.3716	.1782	.1391
	MSE	.2527	1.6012	.0319	.1055
$\beta_1 = 1.00$	Mean	1.0281	.7128	.9968	.9313
	SD	.1466	.1185	.0678	.0711
	MSE	.0221	.0964	.0046	.0098
$\beta_2 = -1.00$	Mean	−1.0240	−.0619	−.9967	−.5226
	SD	.9365	.6144	.2892	.2697
	MSE	.8730	1.2561	.0831	.3002
$\beta_3 = .00$	Mean	.0134	−.2255	.0082	−.1199
	SD	.2662	.1738	.1042	.1010
	MSE	.0707	.0809	.0108	.0264
$\psi = .04$	Mean	.0367	.0648	.0360	.0391
	SD	.0143	.0469	.0097	.0150
	MSE	.0004	.0028	.0001	.0004

*Equivalent to CGEE1 and PQL when analyzed assuming model (21).

level of the within-cluster covariate, $t_{ij} = j - 4$ ($j = 1, \ldots, 7$). Just as Breslow and Clayton observed decreases in the average bias of the PQL regression coefficients with increases in the denominator of the binomial observations, so too do we find that the average bias of the PQL estimator decreases as the number of conditionally independent binary observations increases (i.e., as p_i increases).

8. EXAMPLE

We applied Poisson mixed-effects regression to a set of epileptic seizure count data collected longitudinally over time. The data, as described by Thall and Vail (1990), consist of the number of epileptic seizures counted in 2-week intervals among 59 patients. The patients were randomized to receive a control (placebo) and test drug (progabide) in a two-period crossover design. For simplicity, we follow the analyses of Thall and Vail (1990) and Breslow and Clayton (1993) and examine only the data from the first period. Baseline covariates in the study included (1) $x_{i1} = \log(\text{base}_i)$ which is the baseline seizure counts expressed as the logarithm of 1/4 of an 8-week baseline count; (2) $x_{i2} = \log(\text{age}_i)$, which is the log of the ith patient's age at entry; and (3) G_i, a treatment group indicator variable defined to be 0 for the control drug and 1 for progabide. Included in the model was the time-dependent covariate, visit_{ij} ($j = 1, 2, 3, 4$) representing the four biweekly visits made by each patient and coded as $-3, -1, 1, 3$.

We fit the data assuming the conditional mean, $E(y_{ij}|b_i) = \mu_{ij}(\beta, b_i)$, for the ith subject on the jth visit can be expressed in terms of the log mean counts,

$$\log(\mu_{ij}) = \beta_0 + b_i + \beta_1 x_{i1} + \beta_2 G_i + \beta_3 x_{i1} \times G_i + \beta_4 x_{i2} + \beta_5 \text{visit}_{ij}/10, \quad (22)$$

where $b_i \sim N(0, \psi)$. We also fit the data assuming two different conditional variance-covariance structures. In the first case (model A), we fit the data assuming conditional independence. Here the counts are assumed to be Poisson distributed with mean and variance defined by (22). In the second case (model B), we modeled possible overdispersion by assuming the conditional variance to be $\text{var}(y_{ij}|b_i) = \mu_{ij}(1 + \gamma\mu_{ij})$. This form of overdispersion coincides with the variance of a negative binomial distribution and closely mirrors the unit-level random-effects structure found in model III of Breslow and Clayton (1993). In addition, based on evidence from a preliminary analysis of lagged residuals, we included a "working" first-order autocorrelation structure. Consequently, we fit the data in the second case (model B) assuming a conditional variance-covariance structure of the form

$$\text{var}(y_{ij}|b_i) = \mu_{ij}(1 + \gamma\mu_{ij}), \qquad \text{corr}(y_{ij}, y_{ij'}|b_i) = \rho^{|j-j'|}. \quad (23)$$

Under model A, we fit the data using both CGEE2 and ML estimates with CGEE2 being equivalent to PQL in this setting. The ML estimates were obtained via numerical integration using adaptive Gaussian quadrature and was implemented using the SAS procedure NLMIXED. In the second case (model B), we fit the data using CGEE2. (Here ML is not straightforward given the assumed variance structure.) We did so by specifying the "working" covariance structure (12) where $\boldsymbol{\alpha}' = (\alpha_1, \alpha_2) = (\gamma, \rho)$. The first four moments correspond to those of an overdispersed Poisson random variable with conditional mean μ_{ij}, conditional variance $\mu_{ij}(1 + \gamma\mu_{ij})$, and "working" AR(1) correlation structure $\mathbf{R}_i(\alpha_2)$, with autocorrelation defined by (23). For both cases, we also fit the data using PELS assuming only that the first two moments are defined by (22) and (23).

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Table 3. CGEE2, PELS, and MLE for Thall and Vail's Epilepsy Data ($\hat{\beta} \pm SE$)

	Conditional variance-covariance structure				
	Model A method			Model B method	
Fixed Effects	*CGEE2*[a]	*PELS*[b]	*ML*	*CGEE2*	*PELS*[b]
Intercept	-1.30 ± 1.22	-1.17 ± 1.25	-1.39 ± 1.18	-1.09 ± 1.28	-1.07 ± 1.24
Base (x_{i1})	$.87 \pm .14$	$.93 \pm .14$	$.88 \pm .13$	$.88 \pm .14$	$.92 \pm .14$
Trt (G_i)	$-.92 \pm .41$	$-.89 \pm .42$	$-.93 \pm .40$	$-.89 \pm .43$	$-.85 \pm .41$
Base × Trt	$.33 \pm .21$	$.31 \pm .21$	$.34 \pm .20$	$.31 \pm .22$	$.28 \pm .21$
Age (x_{i2})	$.47 \pm .36$	$.41 \pm .37$	$.48 \pm .35$	$.40 \pm .38$	$.37 \pm .36$
Visit/10	$-.29 \pm .10$	$-.32 \pm .10$	$-.29 \pm .10$	$-.32 \pm .13$	$-.34 \pm .11$
Overdispersion					
γ				$.073 \pm .020$	$.068 \pm .020$
ρ				$-.175 \pm .073$	$-.326 \pm .073$
Random Effects					
ψ	$.27 \pm .06$	$.30 \pm .07$	$.25 \pm .06$	$.31 \pm .07$	$.31 \pm .07$

[a]Equivalent to PQL in this case.
[b]PELS ≡ CGEE2 assuming normality.

Results of these analyses are shown in Table 3. For Model A, the CGEE2 and ML procedures yield very similar estimates as might be expected on the basis of the asymptotic properties of CGEE2. Minimizing the PELS objective function yields estimates that are in surprisingly good agreement with the CGEE2 and ML estimates. For the assumed covariance structure of model B, the CGEE2 and PELS procedures produced very similar estimates of the fixed-effects parameters. Moreover, with the exception of the intercept term, these estimates were not too dissimilar from those obtained under model A. The fact that the estimated lag-one autocorrelation parameter was negative did not come as a surprise in view of the negative correlation that we observed in preliminary plots of the lagged residuals. However, there was evidence that the negative autocorrelation may be due to a few outlying values, a feature worthy of further investigation.

9. DISCUSSION

A number of methods have been proposed for estimating the fixed- and random-effects parameters of a generalized linear or nonlinear mixed-effects model. These include "exact" methods based on numerical integration techniques (i.e., ML) and approximate methods based on conditional first-order Taylor series expansion techniques (i.e., CGEE1). Although simulations suggest that both approaches work well, there will be numerous applications where either ML cannot be readily applied (e.g., model B of our example) or CGEE1 will fail to fully use information about both fixed, and random-effects parameters that appear in the conditional variance. The latter will occur, for example, whenever the underlying distribution is from the quadratic exponential family.

In this article we consider a class of generalized linear and nonlinear mixed-effects models in which the second-order moments are allowed to depend, quite generally, on both fixed- and random-effects parameters. We propose a set of conditional second-order generalized estimating equations (CGEE2) for jointly estimating the fixed- and random-effects parameters assuming that the underlying conditional distribution is a member of the quadratic exponential family. In addition, we show how CGEE2 may also be used to minimize a penalized extended least squares objective function independent of any distribution assumptions. In this case, one need specify only the first two conditional moments. In either case, CGEE2 can be used to simultaneously fit both continuous and discrete data within a general multivariate mixed-effects framework. To illustrate, one could use CGEE2 to estimate the parameters of a shared parameter model for modeling nonignorable missing data (Follmann and Wu 1995). Under the shared parameter model, serial outcome measurements and missing-data indicator variables are assumed to be conditionally independent for a given set of random effects. Consequently, joint estimation of the model parameters can be carried out directly using CGEE2. This bypasses any additional approximations one might otherwise use to estimate the model parameters (e.g., Follmann and Wu 1995).

The CGEE2 algorithm extends Lindstrom and Bates's NLME algorithm and Breslow and Clayton's PQL algorithm to a broader class of conditional distributions. It is equivalent to PQL whenever the conditional distribution is from the linear exponential family. In fact, one can easily verify from (4) and (5) the various conditions under which CGEE2 will reduce to either GEE1, GEE2, or CGEE1. For example, CGEE2 will be equivalent to GEE whenever $\mathbf{b}_i = \mathbf{0}$, $\boldsymbol{\gamma}_i = \mathbf{0}$, and $\boldsymbol{\alpha} = \mathbf{0}$, whereas it will be equivalent to GEE2 whenever $\mathbf{b}_i = \mathbf{0}$ and $\boldsymbol{\gamma}_i \neq \mathbf{0}$. Moreover, it may be possible to apply CGEE2 to cases where the random effects are not normally distributed, thus extending the work of Lee and Nelder (1996) to an even broader class of mixed-effects models.

Under suitable regularity conditions, the CGEE2 estimator is shown to be consistent and asymptotically equivalent to the ML estimator provided that the number of observations per cluster (or subject) goes to infinity. This theoretical result is supported by simulation studies presented here and also in studies reported by Ten Have and Localio (1999). In the latter, empirical Bayes estimates of the random effects using PQL are shown to have minimal bias and adequate confidence interval coverage provided that large cluster sizes are available. Similar results are presented here, although in those cases where

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the underlying distribution is from the quadratic exponential family (e.g., the Gaussian nonlinear random-effects model (18)), the empirical Bayes estimates of the random effects using CGEE2 are shown to outperform similar estimates using CGEE1. The reason, as noted earlier, rests with the fact that CGEE2 uses all relevant information from both the first- and second-order conditional moments.

Although there are a number of advantages associated with CGEE2, there are also several drawbacks worth noting. First and foremost is the potential for biased and inconsistent estimates whenever the conditional second-order moments are misspecified. Indeed, just as population-averaged estimates obtained using GEE2 are sensitive to misspecification of the marginal variance (e.g., Prentice and Zhao 1991; Vonesh and Chinchilli 1997), so too will subject-specific estimates obtained using CGEE2 be sensitive to misspecification of the conditional variance. Moreover, CGEE2 also requires correct specification of the third- and fourth-order moments if it is to achieve asymptotic efficiency. It is not clear what loss in efficiency one can expect when the third- and fourth-order moments are misspecified, and further research along these lines is needed. Finally, a major criticism of CGEE2 (and also of CGEE1/PQL) is that reasonably well-behaved estimates of both fixed- and random-effects parameters are possible only as one increases both the number of subjects and the number of observations per subject. However, such criticism is unwarranted if one is truly interested in estimating both fixed- and random-effects parameters. In the presence of sparse intrasubject data, we know of no methodology, including the computationally more demanding numerical integration techniques, that will lead to well-behaved estimates of the subject-specific parameters. Indeed, subject-specific inference necessarily requires "large" numbers of observations per subject. On the other hand, if one is interested solely in predicting the average individual's response (i.e., estimating the fixed-effects parameters), then numerical integration may be needed to overcome small sample bias (i.e., small p_i) associated with CGEE2. Alternatively, one may be able to improve the performance of CGEE2 by using a bias correction procedure like that of Lin and Breslow (1996). Note, however, that even numerical integration presupposes the underlying model has been correctly specified, an assumption that is not easily verified in the presence of sparse intrasubject data.

In conclusion, although CGEE2 does have limitations, its ease of use and the fact that it is consistent and asymptotically normal under fairly mild regularity conditions should prove useful to researchers from various disciplines. It should also prove useful to those researchers who seek good starting values for more computer-intensive methods.

APPENDIX: ASYMPTOTIC RESULTS

Let $\boldsymbol{\tau}' = (\boldsymbol{\beta}'\boldsymbol{\alpha}') \in \boldsymbol{\Theta}$ and $l_i(\boldsymbol{\tau}, \mathbf{b}_i) = l_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \mathbf{b}_i;\ \mathbf{y}_i) = \log[\pi(\mathbf{y}_i|\mathbf{b}_i, \boldsymbol{\tau}, \boldsymbol{\eta})]$, where $\boldsymbol{\eta}$ denotes a set nuisance parameters associated with third- and fourth-order conditional moments. Let $p_i L_i(\boldsymbol{\tau}, \mathbf{b}_i) = l_i(\boldsymbol{\tau}, \mathbf{b}_i) - \frac{1}{2}\mathbf{b}_i'\boldsymbol{\Psi}^{-1}\mathbf{b}_i$ and let $\hat{\mathbf{b}}_i(\boldsymbol{\tau})$ denote the posterior mode obtained by maximizing $p_i L_i(\boldsymbol{\tau}, \mathbf{b}_i)$ such that

$$\frac{\partial}{\partial \mathbf{b}_i} p_i L_i(\boldsymbol{\tau}, \mathbf{b}_i)\Big|_{\mathbf{b}_i = \hat{\mathbf{b}}_i(\boldsymbol{\tau})} = 0. \tag{A.1}$$

Let

$$l'_{i,b}(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) = \frac{\partial}{\partial \mathbf{b}_i} l_i(\boldsymbol{\tau}, \mathbf{b}_i)\Big|_{\mathbf{b}_i = \hat{\mathbf{b}}_i(\boldsymbol{\tau})},$$

$$l''_{i,bb}(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) = \frac{\partial^2}{\partial \mathbf{b}_i \partial \mathbf{b}_i'} l_i(\boldsymbol{\tau}, \mathbf{b}_i)\Big|_{\mathbf{b}_i = \hat{\mathbf{b}}_i(\boldsymbol{\tau})},$$

$$l'_{i,\tau}(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) = \frac{\partial}{\partial \boldsymbol{\tau}} l_i(\boldsymbol{\tau}, \mathbf{b}_i)\Big|_{\mathbf{b}_i = \hat{\mathbf{b}}_i(\boldsymbol{\tau})},$$

$$l''_{i,\tau\tau}(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) = \frac{\partial^2}{\partial \boldsymbol{\tau} \partial \boldsymbol{\tau}'} l_i(\boldsymbol{\tau}, \mathbf{b}_i)\Big|_{\mathbf{b}_i = \hat{\mathbf{b}}_i(\boldsymbol{\tau})},$$

$$l''_{i,b\tau}(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) = \frac{\partial^2}{\partial \mathbf{b}_i \partial \boldsymbol{\tau}'} l_i(\boldsymbol{\tau}, \mathbf{b}_i)\Big|_{\mathbf{b}_i = \hat{\mathbf{b}}_i(\boldsymbol{\tau})},$$

and

$$l''_{i,\tau b}(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) = \frac{\partial^2}{\partial \boldsymbol{\tau} \partial \mathbf{b}_i'} l_i(\boldsymbol{\tau}, \mathbf{b}_i)\Big|_{\mathbf{b}_i = \hat{\mathbf{b}}_i(\boldsymbol{\tau})}.$$

Similarly,

$$L'_{i,b}(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) = \frac{\partial}{\partial \mathbf{b}_i} L_i(\boldsymbol{\tau}, \mathbf{b}_i)\Big|_{\mathbf{b}_i = \hat{\mathbf{b}}_i(\boldsymbol{\tau})},$$

$$L''_{i,bb}(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) = \frac{\partial^2}{\partial \mathbf{b}_i \partial \mathbf{b}_i'} L_i(\boldsymbol{\tau}, \mathbf{b}_i)\Big|_{\mathbf{b}_i = \hat{\mathbf{b}}_i(\boldsymbol{\tau})}, \text{ etc.,}$$

and

$$l'_{i,b}(\boldsymbol{\tau}, \mathbf{b}_i) = \frac{\partial}{\partial \mathbf{b}_i} l_i(\boldsymbol{\tau}, \mathbf{b}_i),\ l''_{i,bb}(\boldsymbol{\tau}, \mathbf{b}_i) = \frac{\partial^2}{\partial \mathbf{b}_i \partial \mathbf{b}_i'} l_i(\boldsymbol{\tau}, \mathbf{b}_i), \text{ etc.}$$

Also, we denote convergence in probability as $p_i \to \infty$ by $o_p(1_{p_i})$, convergence in probability as $n \to \infty$ by $o_p(1_n)$, and convergence in probability as both $p_i \to \infty$ and $n \to \infty$ by $o_p(1_{p_i,n})$. We prove consistency and asymptotic normality under the following regularity conditions. An outline of the proof when some of these conditions are relaxed is provided at the end.

R1. $p_i = O(p)$ uniformly for $i = 1, \ldots, n$, where $p = \min(p_i)$.

R2. Conditional on $\mathbf{b}_i$, the $y_{i1}, \ldots, y_{ip_i}$ are independently distributed with probability density functions, $\pi(y_{ij}|\mathbf{b}_i, \boldsymbol{\tau}, \boldsymbol{\eta})$, satisfying the necessary regularity conditions (e.g., Bradley and Gart 1962) such that, for fixed $\boldsymbol{\tau}$ and $\boldsymbol{\eta}$, the ML estimate of $\mathbf{b}_i$ is $\sqrt{p_i}$-consistent for $\mathbf{b}_i$ as $p_i \to \infty$. In addition, the necessary regularity conditions are assumed (e.g., Serfling 1980, p. 27, thm. C) such that, by the weak law of large numbers, the following hold true:

$$-p_i^{-1}\frac{\partial^2}{\partial \mathbf{b}_i \partial \mathbf{b}_i'} l_i(\boldsymbol{\tau}, \mathbf{b}_i) = p_i^{-1}\mathbf{Z}_i'\mathbf{V}_i^{-1}\mathbf{Z}_i + o_p(1_{p_i}) \quad \text{as } p_i \to \infty,$$

$$-p_i^{-1}\frac{\partial^2}{\partial \boldsymbol{\tau} \partial \boldsymbol{\tau}'} l_i(\boldsymbol{\tau}, \mathbf{b}_i) = p_i^{-1}\mathbf{X}_i'\mathbf{V}_i^{-1}\mathbf{X}_i + o_p(1_{p_i}) \quad \text{as } p_i \to \infty,$$

$$-p_i^{-1}\frac{\partial^2}{\partial \boldsymbol{\tau} \partial \mathbf{b}_i'} l_i(\boldsymbol{\tau}, \mathbf{b}_i) = p_i^{-1}\mathbf{X}_i'\mathbf{V}_i^{-1}\mathbf{Z}_i + o_p(1_{p_i}) \quad \text{as } p_i \to \infty,$$

and

$$-p_i^{-1}\frac{\partial^2}{\partial \mathbf{b}_i \partial \boldsymbol{\tau}'} l_i(\boldsymbol{\tau}, \mathbf{b}_i) = p_i^{-1}\mathbf{Z}_i'\mathbf{V}_i^{-1}\mathbf{X}_i + o_p(1_{p_i}) \quad \text{as } p_i \to \infty,$$

where, under model (2),

$$E_{y|b}\left\{-\frac{\partial^2}{\partial \mathbf{b}_i \partial \mathbf{b}_i'} l_i(\boldsymbol{\tau}, \mathbf{b}_i)\right\} = \mathbf{Z}_i'\mathbf{V}_i^{-1}\mathbf{Z}_i,$$

$$E_{y|b}\left\{-\frac{\partial^2}{\partial \boldsymbol{\tau} \partial \boldsymbol{\tau}'} l_i(\boldsymbol{\tau}, \mathbf{b}_i)\right\} = \mathbf{X}_i'\mathbf{V}_i^{-1}\mathbf{X}_i, \text{ etc.}$$

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Finally, the matrices $p_i^{-1}\mathbf{Z}_i'\mathbf{V}_i^{-1}\mathbf{Z}_i$ and $p_i^{-1}\mathbf{X}_i'\mathbf{V}_i^{-1}\mathbf{X}_i$ are both assumed to be positive definite with finite determinants such that, for example, the smallest eigenvalue of $p_i^{-1}\mathbf{Z}_i'\mathbf{V}_i^{-1}\mathbf{Z}_i$ exceeds λ_0 for some $\lambda_0 > 0$.

R3. The parameters $\boldsymbol{\eta}$ and $\boldsymbol{\Psi}$ are fixed and known, and the true parameter $\boldsymbol{\tau}^0$ is in the interior of $\boldsymbol{\Theta}$.

R4. For all $\boldsymbol{\tau} \in \boldsymbol{\Theta}$ and all $\mathbf{b}_i \in \Re^v$, the function $L_i(\boldsymbol{\tau}, \mathbf{b}_i)$ is six times differentiable and continuous in $\boldsymbol{\tau}$ and $\mathbf{b}_i$ for all y_{ij}, and $L_i(\boldsymbol{\tau}, \mathbf{b}_i)$ satisfies the necessary regularity conditions needed to change the order of integration and differentiation, as indicated in the proof.

R5. For any $\boldsymbol{\tau} \in \boldsymbol{\Theta}$, there exist $d_1 > 0$ and $\lambda_1 > 0$ such that

a. For all $\boldsymbol{\tau}^* \in B_{d_1}(\boldsymbol{\tau})$, where $B_{d_1}(\boldsymbol{\tau})$ is the $(s+u)$-dimensional sphere centered at $\boldsymbol{\tau}$ with radius d_1, the following holds true:

$$-\frac{1}{n}\sum_{i=1}^{n}\frac{\partial}{\partial\boldsymbol{\tau}'}l_{i,\tau}'(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau}))\Big|_{\boldsymbol{\tau}=\boldsymbol{\tau}^*} = \boldsymbol{\Omega}(\boldsymbol{\tau}^*)^{-1} + o_p(1_n), \quad \text{as } n\to\infty,$$

where $\boldsymbol{\Omega}(\boldsymbol{\tau}^*)^{-1}$ is positive definite with minimum eigenvalue greater than λ_1 and

$$\begin{aligned}\frac{\partial}{\partial\boldsymbol{\tau}'}l_{i,\tau}'(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau}))\Big|_{\boldsymbol{\tau}=\boldsymbol{\tau}^*} &= \{l_{i,\tau\tau}''(\boldsymbol{\tau}^*,\hat{\mathbf{b}}_i(\boldsymbol{\tau}^*)) + l_{i,\tau b}''(\boldsymbol{\tau}^*,\hat{\mathbf{b}}_i(\boldsymbol{\tau}^*))\\ &\quad\times\left[-l_{i,bb}''(\boldsymbol{\tau}^*,\hat{\mathbf{b}}_i(\boldsymbol{\tau}^*)) + \boldsymbol{\Psi}^{-1}\right]^{-1} l_{i,b\tau}''(\boldsymbol{\tau}^*,\hat{\mathbf{b}}_i(\boldsymbol{\tau}^*))\}.\end{aligned}$$

b. The first, second, and up to six derivatives of $\sqrt{p_i}L_i(\boldsymbol{\tau}, \mathbf{b}_i)$ with respect to $\mathbf{b}_i$ are uniformly bounded in $B_{d_1}(\boldsymbol{\tau})$.

R6. At the true value $\boldsymbol{\tau}^0$, the following hold true:

$$E_b\{\mathbf{X}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{X}_i\} = \boldsymbol{\varphi}_i(\boldsymbol{\tau}^0) \quad \text{exists for all } i = 1,\ldots,n,$$

$$\lim_{n\to\infty} n^{-2}\sum_{i=1}^{n}\operatorname{var}_b\{\mathbf{X}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{X}_i\} = 0,$$

and

$$\lim_{n\to\infty} n^{-1}\sum_{i=1}^{n}\boldsymbol{\varphi}_i(\boldsymbol{\tau}^0) = \boldsymbol{\Omega}(\boldsymbol{\tau}^0)^{-1},$$

where $\mathbf{X}_i$, $\mathbf{Z}_i$, and $\boldsymbol{\Sigma}_i$ are evaluated at $\boldsymbol{\tau}^0$ and $\mathbf{b}_i$ and where $\boldsymbol{\Omega}(\boldsymbol{\tau}^0)^{-1}$ is positive definite.

R7. The marginal densities, $\int \pi(\mathbf{y}_i|\mathbf{b}_i,\boldsymbol{\tau},\boldsymbol{\gamma})\pi(\mathbf{b}_i|\boldsymbol{\theta})\,d\mathbf{b}_i$, satisfy the necessary regularity conditions such that the ML estimate of $\boldsymbol{\tau}$ exists and satisfies $(\hat{\boldsymbol{\tau}}_{\mathrm{MLE}} - \boldsymbol{\tau}) = O_p(n^{-1/2})$.

To prove the consistency of the CGEE2 estimator, we first note that for fixed $\boldsymbol{\eta}$ and $\boldsymbol{\Psi}$, the ML estimate of $\boldsymbol{\tau}$ will satisfy the set of estimating equations

$$J(\boldsymbol{\tau}) = \frac{\partial}{\partial\boldsymbol{\tau}}\prod_{i=1}^{n}\pi(\mathbf{y}_i|\boldsymbol{\tau}) = \frac{\partial}{\partial\boldsymbol{\tau}}\prod_{i=1}^{n}\int\exp\{p_iL_i(\boldsymbol{\tau},\mathbf{b}_i)\}\,d\mathbf{b}_i = \mathbf{0}.$$

Under R4, we have

$$\begin{aligned}J(\boldsymbol{\tau}) &= \int\cdots\int\left\{\sum_{i=1}^{n}p_i\frac{\partial}{\partial\boldsymbol{\tau}}L_i(\boldsymbol{\tau},\mathbf{b}_i)\right\}\exp\left\{\sum_{j=1}^{n}p_jL_j(\boldsymbol{\tau},\mathbf{b}_j)\right\}d\mathbf{b}_1,\ldots,d\mathbf{b}_n\\ &= \sum_{i=1}^{n}\sqrt{p_i}\int\cdots\int\left(\sqrt{p_i}\frac{\partial}{\partial\boldsymbol{\tau}}L_i(\boldsymbol{\tau},\mathbf{b}_i)\right)\\ &\quad\times\exp\left\{\sum_{j=1}^{n}p_jL_j(\boldsymbol{\tau},\mathbf{b}_j)\right\}d\mathbf{b}_1,\ldots,d\mathbf{b}_n\\ &= \sum_{i=1}^{n}\sqrt{p_i}\left[\int\left(\sqrt{p_i}\frac{\partial}{\partial\boldsymbol{\tau}}L_i(\boldsymbol{\tau},\mathbf{b}_i)\right)\exp\{p_iL_i(\boldsymbol{\tau},\mathbf{b}_i)\}\,d\mathbf{b}_i\right.\\ &\qquad\left.\times\prod_{j\neq i}^{n}\int\exp\{p_jL_j(\boldsymbol{\tau},\mathbf{b}_j)\}\,d\mathbf{b}_j\right].\end{aligned}$$

Given R2, it follows that, conditional on $\mathbf{b}_i$,

$$L_{i,\tau}'(\boldsymbol{\tau},\mathbf{b}_i) = p_i^{-1}l_{i,\tau}'(\boldsymbol{\tau},\mathbf{b}_i) = p_i^{-1}\mathbf{X}_i'\mathbf{V}_i^{-1}(\mathbf{Y}_i - \mathbf{F}_i) = O_p(p_i^{-1/2}).$$

Furthermore, under R2 it can be shown that the posterior mode satisfies $\hat{\mathbf{b}}_i(\boldsymbol{\tau}) = \mathbf{b}_i + O_p(p_i^{-1/2})$, from which it can be shown that

$$L_{i,\tau}'(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau})) = p_i^{-1}l_{i,\tau}'(\boldsymbol{\tau},\hat{b}_i(\boldsymbol{\tau})) = O_p(p_i^{-1/2}). \tag{A.2}$$

Now, by direct application of the Laplace approximation to integrals of the form $\int\exp\{mh(x)\}\,dx$ and $\int g(x)\exp\{mh(x)\}\,dx$, where $g(x)$ and $h(x)$ are smooth functions in x with $h(x)$ having a unique maximum at some point $\hat{x}$ (see, e.g., Barndorff-Nielson and Cox 1989), it can be shown that

$$\int\exp\{p_iL_i(\boldsymbol{\tau},\mathbf{b}_i)\}\,d\mathbf{b}_i = \exp\{p_iL_i(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau}))\}\left(\frac{2\pi}{|p_i\widehat{L}_{i,bb}''|}\right)^{v/2}(1+O(p_i^{-1}))$$

and

$$\int\left(\sqrt{p_i}\frac{\partial}{\partial\boldsymbol{\tau}}L_i(\boldsymbol{\tau},\mathbf{b}_i)\right)\exp\{p_iL_i(\boldsymbol{\tau},\mathbf{b}_i)\}\,d\mathbf{b}_i = \exp\{p_iL_i(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau}))\}\times\left(\frac{2\pi}{|p_i\widehat{L}_{i,bb}''|}\right)^{v/2}\left(\sqrt{p_i}\frac{\partial}{\partial\boldsymbol{\tau}}L_i(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau})) + O(p_i^{-1})\right),$$

where $\widehat{L}_{i,bb}'' = L_{i,bb}''(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau}))$. Because (A.2) implies $\sqrt{p_i}\times L_{i,\tau}'(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau})) = O_p(1)$, it follows that

$$\left(\sqrt{p_i}L_{i,\tau}'(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau})) + O(p_i^{-1})\right)\times\prod_{j\neq i}^{n}(1+O(p_j^{-1})) = \sqrt{p_i}L_{i,\tau}'(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau})) + O(p^{-1}).$$

Hence we have

$$\begin{aligned}J(\boldsymbol{\tau}) &= \sum_{i=1}^{n}\sqrt{p_i}\left[\exp\{p_iL_i(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau}))\}\left(\frac{2\pi}{|p_i\widehat{L}_{i,bb}''|}\right)^{v/2}\right.\\ &\quad\times\left(\sqrt{p_i}L_{i,\tau}'(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau})) + O(p_i^{-1})\right)\times\prod_{j\neq i}^{n}\\ &\quad\left.\times\left\{\exp\{p_jL_j(\boldsymbol{\tau},\hat{\mathbf{b}}_j(\boldsymbol{\tau}))\}\left(\frac{2\pi}{|p_j\widehat{L}_{j,bb}''|}\right)^{v/2}\times(1+O(p_j^{-1}))\right\}\right]\\ &= K(\boldsymbol{\tau},\hat{\mathbf{b}}(\boldsymbol{\tau}))\left[\sum_{i=1}^{n}p_iL_{i,\tau}'(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau})) + nO_p(p^{-1/2})\right],\end{aligned}$$

where $K(\boldsymbol{\tau},\hat{\mathbf{b}}(\boldsymbol{\tau})) = \exp\{\sum_{j=1}^{n}p_jL_j(\boldsymbol{\tau},\hat{\mathbf{b}}_j(\boldsymbol{\tau}))\}\prod_{j=1}^{n}(2\pi/|p_j\widehat{L}_{j,bb}''|)^{v/2}$. Because $K(\boldsymbol{\tau},\hat{\mathbf{b}}(\boldsymbol{\tau})) \neq 0$ for all $\boldsymbol{\tau}\in\boldsymbol{\Theta}$, the ML estimate, $\hat{\boldsymbol{\tau}}_{\mathrm{MLE}}$, satisfies

$$J(\boldsymbol{\tau})\Big|_{\boldsymbol{\tau}=\hat{\boldsymbol{\tau}}_{\mathrm{MLE}}} = J_1(\boldsymbol{\tau},\hat{\mathbf{b}}(\boldsymbol{\tau}))\Big|_{\boldsymbol{\tau}=\hat{\boldsymbol{\tau}}_{\mathrm{MLE}}} + O_p(np^{-1/2}) = 0, \tag{A.3}$$

where $J_1(\boldsymbol{\tau},\hat{\mathbf{b}}(\boldsymbol{\tau})) = \sum_{i=1}^{n}p_iL_{i,\tau}'(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau})) = \sum_{i=1}^{n}l_{i,\tau}'(\boldsymbol{\tau},\hat{\mathbf{b}}_i(\boldsymbol{\tau}))$ is the set of conditional second-order estimating equations (4) for $\boldsymbol{\tau}$. By taking a Taylor series expansion of $J_1(\boldsymbol{\tau},\hat{\mathbf{b}}(\boldsymbol{\tau}))$ about $\hat{\boldsymbol{\tau}}_{\mathrm{MLE}}$ and noting that, from (A.3), $J_1(\hat{\boldsymbol{\tau}}_{\mathrm{MLE}},\hat{\mathbf{b}}(\hat{\boldsymbol{\tau}}_{\mathrm{MLE}})) = O_p(np^{-1/2})$, we have

$$J_1(\boldsymbol{\tau},\hat{\mathbf{b}}(\boldsymbol{\tau})) = O_p(np^{-1/2}) + J_1'(\boldsymbol{\tau}^*,\hat{\mathbf{b}}(\boldsymbol{\tau}^*))(\boldsymbol{\tau} - \hat{\boldsymbol{\tau}}_{\mathrm{MLE}}), \tag{A.4}$$

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where

$$J_1'(\boldsymbol{\tau}^*, \hat{\mathbf{b}}(\boldsymbol{\tau}^*)) = \frac{\partial}{\partial \boldsymbol{\tau}'} J_1(\boldsymbol{\tau}, \hat{\mathbf{b}}(\boldsymbol{\tau}))|_{\boldsymbol{\tau}=\boldsymbol{\tau}^*} = \frac{\partial}{\partial \boldsymbol{\tau}'} \sum_{i=1}^n p_i L_{i,\tau}'(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau}))|_{\boldsymbol{\tau}=\boldsymbol{\tau}^*}$$

and $\boldsymbol{\tau}^*$ on the line segment joining $\boldsymbol{\tau}$ to $\hat{\boldsymbol{\tau}}_{\mathrm{MLE}}$. By applying the chain rule, we have, for any $\boldsymbol{\tau} \in \boldsymbol{\Theta}$,

$$\begin{aligned} J_1'(\boldsymbol{\tau}, \hat{\mathbf{b}}(\boldsymbol{\tau})) &= \sum_{i=1}^n \left\{ \frac{\partial^2}{\partial \boldsymbol{\tau} \partial \boldsymbol{\tau}'} p_i L_i(\boldsymbol{\tau}, \mathbf{b}_i)\Big|_{\mathbf{b}_i=\hat{\mathbf{b}}_i(\boldsymbol{\tau})} \right. \\ &\qquad \left. + \frac{\partial^2}{\partial \boldsymbol{\tau} \partial \mathbf{b}_i'} p_i L_i(\boldsymbol{\tau}, \mathbf{b}_i)\Big|_{\mathbf{b}_i=\hat{\mathbf{b}}_i(\boldsymbol{\tau})} \frac{\partial \hat{\mathbf{b}}_i(\boldsymbol{\tau})}{\partial \boldsymbol{\tau}} \right\} \\ &= \sum_{i=1}^n \left\{ l_{i,\tau\tau}''(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) + l_{i,\tau b}''(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) \frac{\partial \hat{\mathbf{b}}_i(\boldsymbol{\tau})}{\partial \boldsymbol{\tau}} \right\}. \end{aligned} \tag{A.5}$$

By definition, $\hat{\mathbf{b}}_i(\boldsymbol{\tau})$ maximizes $p_i L_i(\boldsymbol{\tau}, \mathbf{b}_i)$, which implies $\hat{\mathbf{b}}_i(\boldsymbol{\tau}) = \boldsymbol{\Psi} l_{i,b}'(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau}))$. Hence, applying the chain rule once again, we have

$$\begin{aligned} \frac{\partial \hat{\mathbf{b}}_i(\boldsymbol{\tau})}{\partial \boldsymbol{\tau}'} &= \frac{\partial}{\partial \boldsymbol{\tau}'} \{ \boldsymbol{\Psi} l_{i,b}'(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) \} \\ &= \boldsymbol{\Psi} \frac{\partial^2}{\partial \mathbf{b}_i \partial \boldsymbol{\tau}'} l_i(\boldsymbol{\tau}, \mathbf{b}_i)\Big|_{\mathbf{b}_i=\hat{\mathbf{b}}_i(\boldsymbol{\tau})} + \boldsymbol{\Psi} \frac{\partial^2}{\partial \mathbf{b}_i \partial \mathbf{b}_i'} l_i(\boldsymbol{\tau}, \mathbf{b}_i)\Big|_{\mathbf{b}_i=\hat{\mathbf{b}}_i(\boldsymbol{\tau})} \frac{\partial \hat{\mathbf{b}}_i(\boldsymbol{\tau})}{\partial \boldsymbol{\tau}} \\ &= \boldsymbol{\Psi} l_{i,b\tau}''(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) + \boldsymbol{\Psi} l_{i,bb}''(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) \frac{\partial \hat{\mathbf{b}}_i(\boldsymbol{\tau})}{\partial \boldsymbol{\tau}}. \end{aligned}$$

Using the identity $l_{i,bb}''(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) = p_i L_{i,bb}''(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) + \boldsymbol{\Psi}^{-1}$ and solving for $\partial \hat{\mathbf{b}}_i(\boldsymbol{\tau})/\partial \boldsymbol{\tau}'$, we have

$$\begin{aligned} \frac{\partial \hat{\mathbf{b}}_i(\boldsymbol{\tau})}{\partial \boldsymbol{\tau}'} &= \{ -p_i L_{i,bb}''(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) \}^{-1} l_{i,b\tau}''(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) \\ &= \{ -l_{i,bb}''(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})) + \boldsymbol{\Psi}^{-1} \}^{-1} l_{i,b\tau}''(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau})). \end{aligned}$$

On substituting this in (A.5), it follows from R1, R2, and R5 that as $n \to \infty$,

$$\begin{aligned} -\frac{1}{n} J_1'(\boldsymbol{\tau}^*, \hat{\mathbf{b}}(\boldsymbol{\tau}^*)) &= -\frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \boldsymbol{\tau}'} l_{i,\tau}'(\boldsymbol{\tau}, \hat{\mathbf{b}}_i(\boldsymbol{\tau}))\Big|_{\boldsymbol{\tau}=\boldsymbol{\tau}^*} \\ &= \boldsymbol{\Omega}(\boldsymbol{\tau}^*)^{-1} + o_p(1). \end{aligned} \tag{A.6}$$

At this point, for ease of notation, we complete the proof assuming that $\boldsymbol{\tau}$ is scalar. (Details for the multiparameter case are available from the authors.) It follows from (A.6) that for sufficiently large n, $-\frac{1}{n} J_1'(\boldsymbol{\tau}^*, \hat{\mathbf{b}}(\boldsymbol{\tau}^*)) > \lambda_1/2$. This, together with (A.4) and the fact that for sufficiently large p, $|O_p(p^{-1/2})| \le \lambda_1 \delta/4$ for any $0 < \delta < d_1$ (d_1 as defined in R5), imply that, conditional on $\mathbf{b} = (\mathbf{b}_1, \ldots, \mathbf{b}_n)$,

$$P_b\left\{ \frac{1}{n} \sum_{i=1}^n p_i \frac{\partial}{\partial \tau} L_i(\hat{\tau}_{\mathrm{MLE}} + \delta, \hat{b}_i(\hat{\tau}_{\mathrm{MLE}} + \delta)) \le -\frac{\lambda_1 \delta}{4} \right\} \to 1 \quad \text{as } n \to \infty, \ p \to \infty$$

and

$$P_b\left\{ \frac{1}{n} \sum_{i=1}^n p_i \frac{\partial}{\partial \tau} L_i(\hat{\tau}_{\mathrm{MLE}} - \delta, \hat{b}_i(\hat{\tau}_{\mathrm{MLE}} - \delta)) \ge \frac{\lambda_1 \delta}{4} \right\} \to 1 \quad \text{as } n \to \infty, \ p \to \infty.$$

Thus, because $\hat{\tau}_{\mathrm{CGEE2}}$ satisfies $J_1(\hat{\tau}_{\mathrm{CGEE2}}, \hat{b}(\hat{\tau}_{\mathrm{CGEE2}})) = 0$, it follows using standard arguments that

$$\lim_{n,p\to\infty} P_b\{ |\hat{\tau}_{\mathrm{CGEE2}} - \hat{\tau}_{\mathrm{MLE}}| < \delta \} = 1.$$

Hence, using the bounded convergence theorem, we have

$$\begin{aligned} &\lim_{n,p\to\infty} P\{ |\hat{\tau}_{\mathrm{CGEE2}} - \hat{\tau}_{\mathrm{MLE}}| < \delta \} \\ &\qquad = E_b\left\{ \lim_{n,p\to\infty} P_b\{ |\hat{\tau}_{\mathrm{CGEE2}} - \hat{\tau}_{\mathrm{MLE}}| < \delta \} \right\} = 1. \end{aligned} \tag{A.7}$$

Because $J_1(\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}}, \hat{\mathbf{b}}(\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}})) = 0$ and $J_1(\hat{\boldsymbol{\tau}}_{\mathrm{MLE}}, \hat{\mathbf{b}}(\hat{\boldsymbol{\tau}}_{\mathrm{MLE}})) = O_p(np^{-1/2})$, it follows from (A.4) that

$$-J_1'(\boldsymbol{\tau}^{**}, \hat{\mathbf{b}}(\boldsymbol{\tau}^{**}))(\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}} - \hat{\boldsymbol{\tau}}_{\mathrm{MLE}}) = O_p(np^{-1/2}),$$

where $\boldsymbol{\tau}^{**}$ is on the line segment joining $\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}}$ to $\hat{\boldsymbol{\tau}}_{\mathrm{MLE}}$. Because from (A.7) and R7, $\boldsymbol{\tau}^{**}$ is in the interior, it follows from R5 that $-J_1'(\boldsymbol{\tau}^{**}, \hat{\mathbf{b}}(\boldsymbol{\tau}^{**})) \ge \frac{n}{2} \boldsymbol{\Omega}(\boldsymbol{\tau}^{**})^{-1}$, where $\boldsymbol{\Omega}(\boldsymbol{\tau}^{**})^{-1}$ is positive definite. Hence the CGEE2 estimator satisfies

$$(\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}} - \hat{\boldsymbol{\tau}}_{\mathrm{MLE}}) = O_p(p^{-1/2}),$$

from which it follows (given R7) that

$$\begin{aligned} \hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}} &= \hat{\boldsymbol{\tau}}_{\mathrm{MLE}} + O_p\{ \min(p_i)^{-1/2} \} \\ &= \boldsymbol{\tau}^0 + O_p\{ \max[n^{-1/2}, \min(p_i)^{-1/2}] \}, \end{aligned}$$

the desired result.

Because $\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}} = \boldsymbol{\tau}^0 + o_p(1_{p,n})$ as $n \to \infty$ and $p \to \infty$, and because $\hat{\mathbf{b}}_i(\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}}) = \hat{\mathbf{b}}_i(\boldsymbol{\tau}^0) + o_p(1_{p,n}) = \mathbf{b}_i + O_p(p_i^{-1/2}) + o_p(1_{p,n})$, it follows from R1, R2, R6, and the weak law of large numbers that for $N = np$,

$$\begin{aligned} &-\frac{1}{N} J_1'(\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}}, \hat{\mathbf{b}}(\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}})) \\ &= -\frac{1}{N} J_1'(\boldsymbol{\tau}^0, \hat{\mathbf{b}}(\boldsymbol{\tau}^0)) + o_p(1_{p,n}) \\ &= -\frac{1}{n} \sum_{i=1}^n p^{-1} \{ l_{i,\tau\tau}''(\boldsymbol{\tau}^0, \mathbf{b}_i) + l_{i,\tau b}''(\boldsymbol{\tau}^0, \mathbf{b}_i) \\ &\quad \times [-l_{i,bb}''(\boldsymbol{\tau}^0, \mathbf{b}_i) + \boldsymbol{\Psi}^{-1}]^{-1} l_{i,b\tau}''(\boldsymbol{\tau}^0, \mathbf{b}_i) + O_p(p_i^{-1/2}) \} + o_p(1_{p,n}) \\ &= \frac{1}{n} \sum_{i=1}^n \{ p^{-1} \mathbf{X}_i' \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i + o_p(1_p) \} + o_p(1_{p,n}) \\ &= \{ p^{-1} \boldsymbol{\Omega}(\boldsymbol{\tau}^0)^{-1} + p^{-1} o_p(1_n) \} + o_p(1_p) + o_p(1_{p,n}). \end{aligned}$$

It follows from R6 that $\{ n^{-1} \sum_{i=1}^n p^{-1} \mathbf{X}_i' \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i \}^{-1} = p \boldsymbol{\Omega}(\boldsymbol{\tau}^0) + p o_p(1_n)$, and hence that

$$\left\{ \sum_{i=1}^n \mathbf{X}_i' \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i \right\}^{-1} = \frac{1}{n} \boldsymbol{\Omega}(\boldsymbol{\tau}^0) + o_p(1_n) O(n^{-1}).$$

Thus it can be shown, using Slutsky's theorem, that

$$\sqrt{n}(\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}} - \boldsymbol{\tau}^0) \xrightarrow{d} N(0, \boldsymbol{\Omega}(\boldsymbol{\tau}^0)).$$

One can extend the proof to include conditionally dependent correlated data under some additional regularity conditions on the correlation structure (e.g., Serfling 1980, p. 27, thm. E). Likewise, one can extend the proof to those cases where $\boldsymbol{\eta}$ and $\boldsymbol{\Psi}$ are unknown by replacing them with consistent estimates. Note that at the final estimate, $\hat{\boldsymbol{\tau}}_{\mathrm{CGEE2}}$, the estimate $\widehat{\boldsymbol{\Psi}}$ in (8) can be shown to be consistent for $\boldsymbol{\Psi}$ as $n \to \infty$ and $p \to \infty$ (see, e.g., Demidenko 1997).

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